Frank Rafael Sánchez Monteza

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Education

Department of Physics UPR - Mayagüez Campus — Mayagüez, Puerto Rico
Master of Science in Physics — 2024 - 2026 *(expected)*
- Advisor: Prof. Francisco Bezares Salinas
- Research area: Photonic Nanomaterials

Professional School of Physics, National University of Trujillo — Trujillo, Peru
Bachellor of Science of Physics — 2018 - 2023
- Advisor: Prof. Segundo R. Jáuregui-Rosas
- Research area: Synthesis and Characterization of Nanomaterials

Publications

1. **Frank R. Sánchez-Monteza,** Segundo R. Jáuregui-Rosas, Olga V. Soledad Flores. Physical properties of Mn_3O_4 nanoparticles one-step synthesized at room temperature and short time. *Chemical Engineering Transactions*, 2026.

Projects

Physical Vapor Deposition of metal nanoparticles for the scalable, cost-effective fabrication of novel nanophotonic devices (Master's Thesis Student)
Puerto Rico Science, Technology & Research Trust — 2025/01 - present

Study of alpaca and vicuña superfiber: Nanotechnology-based enhancement for use as a multifunctional material (Member of the Research Seedbed)
Consejo Nacional de Ciencia y Tecnología (CONCYTEC) — 2020/01 - 2022/08

Work Experience

University of Puerto Rico - Mayagüez Campus | Mayagüez, PR — 2024/01 - present
- Laboratory Instructor of Physics

Multidisciplinary research laboratory of nanoscience and nanotechnology "ORSR" - National University of Trujillo | Trujillo, Peru — 2023/07 - 2023/12
- Synthesize and Characterization of nanomaterials
- **Professional practitioner**

Multidisciplinary research laboratory of nanoscience and nanotechnology "ORSR" - National University of Trujillo | Trujillo, Peru — 2022/05 - 2023/01
- Synthesize and Characterization of nanomaterials
- **Preprofessional practitioner**

Paijan Municipal Academy | Paijan, Peru — 2023/02 - 2023/03
- Teaching of the Pre-University Physics course
- **Teacher**

Aiapaec Academy | Trujillo, Peru — 2023/01 - 2023/02
- Teaching of the Pre-University Physics course
- **Teacher**

Research Experience

- **Summer Internship**, NANOSUMMER 2019 - Multidisciplinary research laboratory of nanoscience and nanotechnology "ORSR" — 2019/01 - 2019/02
- **Research Project**, PICFEDU-2022 UNT
  "Influence of Annealing Temperature on the Structural and Optical Properties of TiO_2 Nanoparticles" — 2022/01 - 2022/12
- **Research Project**, CONCYTEC
  "Study of alpaca and vicuña superfiber: Nanotechnology-based enhancement for use as a multifunctional material" — 2020/01 - 2022/08

Professional Skills

Languages : Spanish (Native), English (A2)
Ofimatic : Word, PowerPoint, Excel, Publisher
Canva : For the design of flyers and content creation
Arduino IDE : For scientific instrumentation
OriginPro : For data analysis
VESTA : For crystallography
Tracker : For physics laboratory
FilmoraX : For edit videos
Programming : `Matlab`, `Python`

Courses

Attended

Workshop: Safety and Risk Prevention in Metallurgical and Materials Processes
| CIP CDLL - Perú
- January 17th, 2023, with a duration of 12 hours.

Training: Hands-on Training on Reactor Operations, UF Reactor Sharing Program
| UF - USA
- From September 8th to 12th,2025

Course: Latex
| UNI - Perú
- From July 31th to August 19th,2025, with a duration of 16 hours.

Course: Data Science 1: Exploratory data analysis
| UNI - Perú
- From July 30th to August 11th,2025, with a duration of 16 hours.

Webinar: Tips for Writing a Scientific Article with Nathalie Jacobs
| Springer
- March 13th, 2023

Course: Fundamentals of electrochemical techniques for nanomaterial-based sensors
| UNT - Perú
- From November 4th,2022 to January 13th,2023, with a duration of 40 hours.

Course: Responsible Conduct in Research
| CONCYTEC - Perú
- Approved on December 11th, 2022

Course: Microsoft Office Professional 2019
| CLC - Perú
- 25-week in 2021 Virtual Course (360h)

Training: Multidisciplinary Research: Capabilities and Challenges in Responding to the Global Trend
| UNT - Perú
- From December 8th to 31th, 2021, with a duration of 48h

Course: Structural Characterization by Nuclear Magnetic Resonance
| SMM - Mexico
- From March 19th to 30th, 2021 a duration of 15h

Course: Learning to program in R from scratch: oriented towards research in biological sciences
| Biology - UNT
- March 20th and 21th, 2021, with a duration of 30h

Workshop: Techniques for characterizing biomaterials and biological samples
| National Polytechnic Institute - Mexico
- From March 15th to 19th, 2021

Course: Learning to program in LaTeX from scratch: oriented towards research in biological sciences
| Biology - UNT
- March 6th and 7th a duration of 30h, 2021

Summer Workshop on Gravitation
| Physics - UNT
- From February 3rd to 7th, 2020

Annual Astrobiology Training | SCAP
- From September 15th to November 17th, 2018

Presenter

Workshop "Experimental Physics for Regular Basic Education Teachers"
| UNT - Perú
- From November 30 to December 17, 2022, with a duration of 24 hours.

Academic Events

Presenter

Symposium "Quantum Week UNT - 2025"
| UNT - Perú
- From April 11th to 13th, 2025

Symposium "Quantum Talks in Puerto Rico"
| UPRM - Puerto Rico
- April 7th and 8th, 2025

"II Student Workshop on Computational Physics"
| UNT - Perú
- October 21th and 22th, 2021

Plenary sessions of the Research Seedbed for the Project "Study of Alpaca and Vicuña Superfibers: Nanotechnology Improvement for Use as Multifunctional Materials" under contract No. 48-2019-FONDECYT-BM-INC.INV"
| UNT - Perú
- From January to June, 2021

"5th Basic Sciences and Applications Fair"
| UNT - Perú
- From November 20th to 22th, 2019

"Feria Perú con ciencia descentralizada para la macro Región Norte - EUREKA (CONCYTEC) - Trujillo"
| UNT - Perú
- November 8th, 9th and 10th, 2019

"4th Basic Sciences and Applications Fair"
| UNT - Perú
- From November 5th to 9th, 2018

Attended

Symposium "Academic Conference for the Anniversary of the Installation of the First Faculty Council"
| UNT - Perú
- From October 13th to 15th, 2021

Conferences "Keynote Lectures for the Anniversary of the Installation of the First Faculty Council"
| UNT - Perú
- From October 13th to 15th, 2021

Conferences "1st Academic Meeting of Physics Graduates"
| UNT - Perú
- June 9th and 16th, 2021

Plenary sessions of the Research Seedbed for the Project "Study of Alpaca and Vicuña Superfibers: Nanotechnology Improvement for Use as Multifunctional Materials" under contract No. 48-2019-FONDECYT-BM-INC.INV"
| UNT - Perú
- From January to June, 2021

"Academic Conference for Graduate Physicists 2019 (JAFE-2019)"
| UNT - Perú
- January 7th and 8th, 2020

Organizer

Symposium "Quantum Week UNT"
| UNT - Perú
- April 13th, 14th and 15th, 2023

Member of the organizing committee of the Course "Fundamentals of Electrochemical Techniques for Nanomaterial-Based Sensors"
| UNT - Perú
- From November 4th, 2022 to January 13th, 2023

Member of the 2022 Academic Events Subcommittee for the 60th Anniversary of the Faculty of Physical and Mathematical Sciences"
| UNT - Perú
- Organizer of the 2022 Academic Conference

Organizer of I Academic meeting of physics graduates
| UNT - Perú
- June 9th and 16th, 2021

Organizer and participant of the 2020 Annual Astrobiology Training
| Perú
- From December 6th to 14th, 2020

Groups and organizations

- **Member of research group Nanoscience and Nanotechnology – GMIN - UNT**
  Researcher
- **Member of research group Nanophotonics - UPRM**
  Researcher
- **Meteorological Society of Puerto Rico - UPRM**
  Colaborator
- **University of Puerto Rico, Optics and Photonics - UPRM**
  Member